Curriculum Vitae

Kavya Agrawal, PhD Scholar, TERI School of Advanced Studies, New Delhi
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Education

2024 – present	**PhD, Natural & Applied Sciences** TERI School of Advanced Studies, New Delhi. Coursework CGPA 9.0/10.
2021 – 2023	**MA Geography** Dr. B.R. Ambedkar University, Agra. CGPA 7.60.
2018 – 2021	**BA (Honours) Geography** University of Delhi. CGPA 7.86.
2016 – 2018	**Senior Secondary School** Delhi Public School, Aligarh. CGPA 9.80.
2014 – 2016	**Secondary School** St. Francis Inter College, Hathras. CGPA 9.20.

Doctoral research

Thesis	**Development of an Adaptable Machine Learning Based Framework for Multi-Contaminant Prediction**
Supervisors	**Prof. Chander Kumar Singh (supervisor); Dr. Adwitiya Sinha (co-supervisor)**
Status	**Full-time. Registered 1 February 2024. Comprehensive examination completed 28 January 2026.**
Coursework	**Research and Publication Ethics (RPR275), A; Research Methodology (RPR271), A; Environmental Statistics (NRE115), A; Digital Image Processing & Information Extraction (NRG172), A**

Research experience

2025 – 26	**Uranium in groundwater across India** *Paper submitted to journal; Best Paper Award, ICSSR EHP NEXUS 2026*
2026 –	**Fluoride and the effect of spatial resolution** *In progress*
2026 –	**Downscaling and upscaling environmental predictors** *In progress*
2024 – 25	**Arsenic in the Middle Gangetic Plain** *Presented at As2024, Bhubaneswar*
2023	**Land surface temperature trends across the Indus River Basin** *Published, Journal of Mountain Science*
2023	**Identification and mapping of permafrost in the Indus Basin**
2023	**Snow cover and changes in Indian basins**

Publications

2023 — **Evaluating spatial and elevation-wise daytime/nighttime LST trends across the Indus River Basin**
Mal, S., **Agrawal, K.**, Rani, S. et al. *Journal of Mountain Science* 20, 3154–3172. doi.org/10.1007/s11629-023-8157-8

2023 — **Analyzing the Possibilities and Prospects of 24/7 Water Supply in Delhi, India**
Sanu, S. K., Chandrakanta, **Agrawal, K.**, Sharma, V. R., & Bajaj, A. *Journal of Indian Water Works Association*, LV(1). UGC-CARE Group I, ISSN 0970-275X.

Manuscripts

2025–26 — **Predicting Uranium Occurrences in Groundwater throughout India**
Research paper
Submitted to journal

2026 — **Comparing and Benchmarking Deep Learning Models for Predicting Groundwater Contamination caused by Fluoride with the effects of Spatial Resolution on Prediction**
Research paper
In progress

2026 — **A Physically Informed Framework for Multi-Resolution Downscaling and Upscaling of Environmental Predictors (Satellite Data)**
Research paper
In progress

Conference presentations

Apr 2026 — **Machine-Learning-Based Uranium Risk Assessment, Population Exposure and Groundwater Governance**
ICSSR EHP NEXUS 2026, Track 10: AI and Predictive Analytics, Gautam Buddha University, 27–28 April 2026.
First Position, oral presentation, and Best Paper Award

Oct 2024 — **Distribution, Speciation and Controls on As Mobilization in the Groundwater in Aquifers of the Middle Gangetic Plain, India**
9th International Congress & Exhibition on Arsenic in the Environment (As2024), KIIT Deemed to be University, Bhubaneswar, 20–24 October 2024. Poster.

Nov 2022 — **Problems and Prospects of Delhi's Drinking Water Supply**
IGU Thematic Conference on Sustainability, Future Earth and Humanities, Department of Geography, Central University of Haryana, Mahendragarh, 24–25 November 2022.

Jul 2022 — **Educational Relevance of Ramayana in Modern India**
National Seminar on Promoting Human Values and Culture for Peaceful Co-Existence, Amity University Haryana, sponsored by ICPR, 22 July 2022.

May 2022 — **Digital Enhancement of Physical Space and Connective Turn in Tourism Industry during and Post-COVID-19 Period**
International Conference on Geography and the Digital: Intersections and Conflations, Department of Geography, Delhi School of Economics, University of Delhi, 23–25 May 2022.

May 2022 — **Migrant College Students during Covid Pandemic: Evidence from the Field**
Jamia International Conference on Education 2022, Faculty of Education, Jamia Millia Islamia, New Delhi, 6–8 May 2022.

Apr 2022 — **Migrant College Students during Covid Pandemic: Evidence from the Field**
International Conference on Challenges to Disaster Risk Reduction and Resilient Habitat, Centre for Disaster Management Studies, Shaheed Bhagat Singh College, University of Delhi, 5–6 April 2022.

Teaching

Jun–Dec 2026 — **Principles of GIS and GNSS (NRG176)**
MSc Geoinformatics, TERI School of Advanced Studies. 4 credits, 60 hours (L-T-P 42-2-32).
Course teaching

Jun–Dec 2026 — **Principles of Remote Sensing (NRG169)**
MSc Geoinformatics, TERI School of Advanced Studies. 4 credits, 60 hours (L-T-P 42-2-32).
Course teaching

Jul–Dec 2025 — **Principles of Remote Sensing (NRG169)**
MSc Geoinformatics, TERI School of Advanced Studies. 32 practical learning hours.
Course taught

Projects

Postgraduate, 2021–2023

- Measuring the Problems and Emotions of Urban-Ward Migrants: A Case Study of Construction Workers in Delhi
- Monitoring the Effects of Light Pollution & Evaluating the Effectiveness of Indian Regulations
- Community Awareness, Participation, & CCDRR: Reflections from North-West India
- Experience of Migrant College Students During COVID-19 Pandemic
- Analyzing the Possibilities and Prospects of 24/7 Water Supply in Delhi, India
- Geography of Ramayana
- Digital Enhancement of Physical Space and Connective Turn in the Tourism Industry during and Post-COVID-19 Period

Undergraduate, 2018–2021

- Forest Monitoring & Urban Sprawl Analysis Using GIS in Delhi
- LULC Change Detection for 20 Years in South-West Delhi
- Social Indicators & Status Analysis of Jammu & Kashmir
- Socio-Economic Impact of COVID-19 in Delhi
- Snow Cover & Changes in Indian Basins
- HRVC Analysis of Jim Corbett
- Studying Human Trafficking in Delhi through Remote Sensing
- Understanding Ansal Plaza Mall of South Delhi as a Gendered Landscape: A Temporal Study

Awards and achievements

2026 — **First Position, oral presentation, and Best Paper Award**
ICSSR EHP NEXUS 2026, Gautam Buddha University.

2025 — **2nd Rank, Map Making Competition**
Amity University.

2018 — **State Child Scientist Award**
National Science Congress.

2015 – 2018	**Consecutive top 3 ranks, English Olympiads** British Olympiad Foundation.
2016, 2018	**Merit Awards**
	20 cultural and general knowledge awards From various organisations, including prizes in dance.

Rapporteurship and policy engagement

Mar 2026	**Rapporteur, National Workshop on Geospatial Technology for Climate Action, Sustainability and Disaster Risk Reduction** TERI School of Advanced Studies in collaboration with NRSC, ISRO, 16–20 March 2026.
Feb 2026	**Delegate, World Sustainable Development Summit 2026** Taj Palace, New Delhi, 25–27 February 2026. Transformations: Vision, Voices, and Values for Sustainable Development and Climate Action.
Mar 2025	**Rapporteur, Ministerial Session: NDC 3.0 and Multilateralism on the Road to COP30 and COP33** World Sustainable Development Summit 2025, TERI.
Mar 2025	**Rapporteur, High-Level Plenary on Achieving Sustainable Development through Integrated Action** World Sustainable Development Summit 2025, TERI.

Positions of responsibility

2026	**Facilitator and Judge, Programme LEAD 2026** TERI SAS. Sustainable City Design Challenge for students of classes 9–12.
Nov 2025	**Judge and Panel Member, TERI SAS Sustainability Hackathon 2025** Expert evaluation panel for school and undergraduate sustainability proposals, 18–19 November 2025.
2025	**Judge, KnoxCode 2025, Geospatial Innovation Challenge** Organised by GeoknoX, TERI SAS, as part of Aahwaan Fest 2025.
2018 – 2021	**Editor-in-Chief, Geography Association** University of Delhi.
2018 – 2020	**Magazine Coordinator, *Vani*** Shaheed Bhagat Singh College, University of Delhi.
2019	**English Editor, *Vani*** Shaheed Bhagat Singh College, University of Delhi.

Workshops, seminars and summits attended

Feb 2026	**Water as a Commons workshop, Water Governance Working Group** With the Ostrom Workshop, Indiana University Bloomington and IU India Gateway, 26–27 February 2026; included a Yamuna River field visit with INTACH Delhi.
Feb 2026	**Ambition to Action: Climathon, Delhi Climate Innovation Week** Climate Career & Sustainability Expo, Innovate X Delhi, 21 February 2026.
Feb 2026	**Guest lecture: Soil Carbon Sequestration as a Climate Mitigation Mechanism** Ms. Violeta Martinkė, Project Development Lead, InSoil, 13 February 2026.

Oct 2025	**Act4Earth Dialogue: Strengthening Multilateralism on the Road to COP30 and Beyond** Virtual, 17 October 2025.
Oct 2025	**Expert talk: DHI Urban Water Solutions** Mr. Morten Just Kjølby, Product Portfolio Manager for MIKE+, DHI Denmark, TERI SAS, 7 October 2025.
Sep 2025	**Science Summit 2025: Marine Resources for Sustainable Development** 16 September 2025.
Aug 2025	**QGreenland Educator Workshop** Virtual, 5–7 August 2025. Developing educational activities with QGreenland data.
Jul 2025	**Roots to Resilience Seminar** TERI SAS with UNEP, UNESCO and Akashvani, 25 July 2025.
Mar 2025	**World Sustainable Development Summit 2025** New Delhi, 5–7 March 2025. Partnerships for Accelerating Sustainable Development and Climate Solutions.
Oct 2024	**Lecture by Professor John A. Cherry on groundwater and sustainable development** Lee Kuan Yew Water Prize recipient, 24 October 2024.
	Climate Skills: Seeds for Transition TERI SAS and TERI with British Council and HSBC Bank.

Training, capacity building and schools

Mar 2026	**National Workshop on Geospatial Technology for Climate Action, Sustainability and Disaster Risk Reduction** TERI SAS in collaboration with NRSC, ISRO, 16–20 March 2026. Participant, volunteer and rapporteur.
Dec 2024	**Capacity Building Workshop on UAV/Drone Applications in Agriculture and Natural Resource Management** TERI SAS with IIT BHU, IIT Roorkee, Namami Gange, Hokkaido University, GLP Earth, GLP Japan, Abhipsa Foundation and Kambil, 10–12 December 2024.
Nov – Dec 2024	**21-day Winter School Capacity Building Program in Geospatial Science and Technology** Amity University, Noida, supported by the National Geospatial Program, DST, Government of India, 18 November – 7 December 2024.
Mar 2024	**Harnessing Geospatial Technology for Forest Fire Disaster Management: Mapping, Monitoring and Mitigation** One-week capacity building programme, Amity University, Noida, 11–15 March 2024.

Skills

GIS and remote sensing

Google Earth Engine (JavaScript), ArcGIS / ArcGIS Pro, QGIS, ERDAS Imagine, Google Earth; image enhancement and feature extraction, orthorectification and mosaicking, DEM processing, satellite image acquisition and processing.

Programming and machine learning

Python, pandas; machine learning, deep learning and AutoML (AutoGluon); Random Forest, LightGBM; explainable AI (SHAP, LIME).

Spatial analysis and statistics

Earth observation and environmental data

Spatial data analysis and visualisation; time-series modelling and trend analysis; Mann–Kendall test, Theil–Sen slope; Moran's I, semivariograms.

Parameters

MAAT, BMAT, PISR, land surface temperature, NDVI/NDBI, night-time light, snow, precipitation, slope, aerosols and solar radiation, hydrochemistry (pH, EC, major ions, nitrate, fluoride, uranium).

Research and field

Field and survey-based research, socio-environmental data analysis, drinking-water quality assessment, disaster risk reduction, climate–water–environment interactions, sustainability and climate policy.

Landsat 8/9, Sentinel, MODIS, VIIRS, FIRMS, CHIRPS, IMD, ERA5-Land, TerraClimate, GLDAS, GRACE, SRTM, MERIT Hydro, JRC Global Surface Water, OpenLandMap, GPW, GHM, GHSL, SMOD, LCZ; CGWB hydrochemical observations.

Data quality

Provenance and lineage, duplicate and sentinel-value handling, coordinate validation, unit checks, resolution harmonisation, independent validation, masked-pixel handling.

Professional

Academic writing and conference presentations, scientific communication and reporting, editorial and content development, rapporteurship and policy documentation, academic coordination, interdisciplinary collaboration.

Volunteering and co-curricular

2016 – present	**Volunteer, SWAPO NGO**
2019 – 2021	**National Service Scheme (NSS)**
2018 – 2019	**Coordination of Volunteers, Vasundhara**
2018 – 2023	**Personal blogging and content writing**

Interests

Collecting old coins and stamps, reading historic stories and accounts, photography and videography, travelling, badminton and dance.

Contact

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